Statistical Learning with Extreme Values

Presentation - Paper Reading Assignment

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Sparse representation of multivariate extremes with applications to anomaly detection

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MATHÉMATIQUES VISION APPRENTISSAGE

The "Crypto" Intuition: When is a Crash an Anomaly?

The Scenario: You manage a portfolio of 6 assets: `BTC`, `ETH`, `BNB`, `SOL`, `XRP`, and `USDT` (Stablecoin).

Two types of Extreme Events:

Type A: Systemic Crash

Bitcoin drops 20%, everything else drops 20%.

Is this an anomaly?
NO. It's a standard market crash.

Type B: The De-peg

Bitcoin is stable, but **USDT** (Stablecoin) drops 20%.

Is this an anomaly?
YES. This is structural failure.

The Challenge: Standard anomaly detection (Distance/Correlation) fails here because both events look "far from the mean." We need **Multivariate EVT**.

The Theoretical Barrier: The Curse of Dimensionality

Goal: Model the dependence of extremes (The Angular Measure Φ).

The Problem

- In high dimensions ($d > 10$), the "sphere" of directions is too big.
- Data is sparse: We cannot estimate the density everywhere.

The Solution (Goix et al., 2017) [GSC17]: Sparsity Hypothesis

- Extremes don't happen "everywhere" on the sphere.
- They concentrate on specific low-dimensional **sub-cones** (faces).

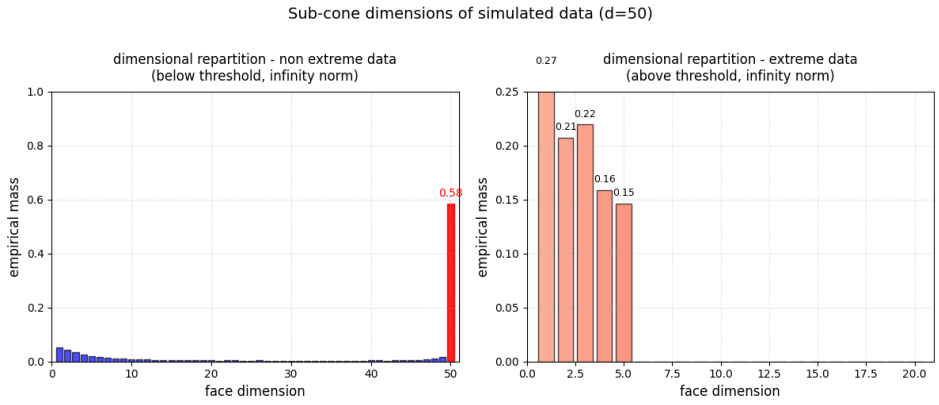
Intuition: In our data, a crash of $\{BTC, ETH\}$ is common.
A crash of $\{BTC, USDT\}$ is physically unlikely.

Geometric Approach: ϵ-Thickened Rectangles

How do we find these "faces" with finite data? We introduce a tolerance ϵ.

An observation x belongs to cone α if:

1. Features in α are **Large** ($> \epsilon ||x||_\infty$)
2. Features NOT in α are **Small** ($\leq \epsilon ||x||_\infty$)



Visualizing mass concentration on low-dim cones

The Estimator: We count how many extreme points fall into each cone.

$$\hat{\mathcal{M}}(\alpha) = \frac{1}{k} \sum_{i=1}^{n} \mathbb{I}\{\text{Point } i \in \text{Cone } \alpha\}$$

DAMEX: Detecting Anomaly with Multivariate EXtremes

1. **Standardize:** Transform data to Pareto scale (Rank transform).
2. **Learn:** Estimate the mass $\hat{\mathcal{M}}(\alpha)$ for every cone (The "Normal" Profile).
3. **Score:** For a new extreme point x:

The Anomaly Score

$$s_n(x) \propto \frac{\text{Mass of the Cone containing } x}{\text{Magnitude of } x}$$

- **Systemic Crash:** $x \in \{BTC, ETH, ...\}$. This cone has **High Mass**. $\Rightarrow$ Score is High (Normal Extreme).
- **De-peg:** $x \in \{BTC, USDT\}$. This cone has **Zero Mass**. $\Rightarrow$ Score is Low (Anomaly).

Results: The Structure of Crypto Crashes

We applied DAMEX to daily returns (2020-2024). **What "Normal" extremes did it find?**

Cone Type	Assets Involved	Mass (Likelihood)
1. Systemic	BTC + ETH + BNB + SOL + XRP *(The whole market moves together)*	**1.00**
2. Idiosyncratic	USDT (Stablecoin) *(Stablecoin volatility is isolated)*	**0.72**
3. Noise	SOL + USDT *(Rare/Weak dependence)*	**0.18**

Finding: The algorithm successfully "decoupled" the Stablecoin from the speculative assets without supervision.

Conclusion

Summary:

1. **Sparsity is key:** In high dimensions, extremes concentrate on low-dimensional faces (sub-groups).
2. **Geometry over Magnitude:** Anomalies in the tail are defined by *directions* (ϵ-cones), not just distance.
3. **Practical Value:** DAMEX allows risk managers to distinguish "Panic Selling" (Systemic) from "Asset Failure" (Structural).

Future Work:

- Selecting ϵ and k is hard. We implemented an AIC criterion (see report) to automate this.

Questions?

Bibliography I

[GSC17] Nicolas Goix, Anne Sabourin, and Stephan Clémençon. "Sparse representation of multivariate extremes with applications to anomaly detection". In: *Journal of Multivariate Analysis* 161 (2017), pp. 12–31. ISSN: 0047-259X. DOI: https://doi.org/10.1016/j.jmva.2017.06.010. URL: https://www.sciencedirect.com/science/article/pii/S0047259X17304062.

Appendix